Algebraic problem solving: Finding the correct approach

1. Find all **integers** x,y and z such that

$$x^2 + y^2 + z^2 = 2(yz + 1)$$

   and

$$x + y + z = 2022.$$

2. Prove that $n^3 - 11n^2 + 36n - 26$ is not prime for any integer value of n.

3. Is the value of $n^2 - 3n + 13$ a prime number for every non-negative integer value of n?

4. Find all the integer values of n for which $9n^2 - 6n - 14$ is an exact square.

5. Solve, for **real values** of $x, y,$ and z, the simultaneous equations:

$$x(y + z) = 102$$
$$y(z + x) = 120$$
$$z(x + y) = 126.$$

6. Find all **real numbers** x such that $(1 + x)^6 - 2(1 - x)^6 = (1 - x^2)^3$.

7. Prove that there is no pair of **positive integers** a and b that satisfy the equation

$$4a(a + 1) = b(b + 1).$$

8. Find all **real solutions** of the simultaneous equations

$$x^2 = 2y + 2z + 5$$
$$y^2 = 2z + 2x + 5$$
$$z^2 = 2x + 2y + 5$$

9. Find all real values of x, y, and z such that $(x + 1)yz = 12$, $(y + 1)zx = 4$, and $(z + 1)xy = 4$.

10. Given any positive integer $m > 1$, is it always possible to find two *distinct* positive integers whose product is m times their sum? Is it ever possible to have two different pairs of positive integers with this property?